

Project 2

Budget-Constrained Stock Selection with 0/1 Knapsack on S&P 500 Data

MSML606

Team Composition: Individual Project

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Statement

- The use of external resources is clearly cited. Please refer to the bold words in my submission.
- If no citation is found for any specific part of my submission, it means "No external sources were used; all ideas are my own or from course lecture slides".
- The core logic and initial approach are manually authored and typed by the author Chenhongshu Yu.

Problem

US stock market is one of the most vibrant market in the world. It is full with opportunities and also the risk. In real investing, even though many stocks look attractive, it is usually not reasonable to buy all of them, because doing so may create a portfolio that is too risky.

This project studies how to choose the best selection (basket) of stocks when an investor cannot take unlimited risk. I model this as a budget-constrained stock selection problem: each stock has a potential benefit and a risk cost, and the goal is to select the combination of stocks that gives the highest total benefit while staying under a fixed risk budget.

To be more specific, each stock will be treated as one item in a **0/1 knapsack problem**. For the simple measurement of the risk, the key points are set as:

- each stock = one item
- value = a momentum-based score derived from recent price performance
- weight = the stock's 20-day volatility
- capacity = fixed total risk budget

The Question:

Which subset of stocks should be selected to maximize total attractiveness while keeping total risk under control?

Dataset

Kaggle S&P 500 dataset: <https://www.kaggle.com/datasets/jacksaleeby/s-and-p500-historical-data>

I choose a public Kaggle dataset containing daily OHLCV stock data for current S&P 500 companies. The dataset includes historical daily prices and trading volume for 472 current S&P 500 companies, with more than 2.7 million rows of data spanning January 2000 to February 2026. This dataset is suitable because it provides the price history needed to compute momentum-based scores and rolling volatility measures for each stock.

From the dataset, I will use daily closing prices and possibly trading volume to compute features such as recent returns and 20-day volatility. These features will then be converted into the value and weight inputs required by the optimization algorithm.

Algorithm and Usage

The main algorithm will be **0/1 knapsack** solved with **dynamic programming (DP)**. For each decision date, every stock is represented by:

- value = a momentum-based score derived from recent price performance
- weight = the stock's 20-day volatility
- capacity = the fixed maximum total risk budget

The dynamic programming algorithm will search for the subset of stocks that maximizes the total value while ensuring that the sum of their risk weights stays below a fixed risk budget. This gives an exact solution under the project's simplified portfolio model.

As a comparison baseline, I will also implement a **greedy method** that ranks stocks by their value-to-risk ratio and selects them in that order until the risk budget is reached. This baseline is faster and easier to explain, but it may not always find the optimal portfolio. Comparing the greedy approach with dynamic programming will show the benefit of using a more principled algorithms method.

FallBack Plan

If the stock-selection project becomes too difficult to implement within the project timeline, the backup project will be **flight route planning using graph algorithms**. In this version, airports are modeled as nodes and direct flights as edges in a graph, and algorithms such as **BFS** and **Dijkstra's algorithm** can be used to find routes with the fewest stops or the

shortest total distance. This fallback still uses a real-world dataset from kaggle:
<https://www.kaggle.com/datasets/elmoallistair/airlines-airport-and-routes>